

Gaussian Process I

Def: $\forall n, \forall t_1, \dots, t_n, (x(t_1), \dots, x(t_n))^T = X$, 有 $X \sim N(\mu, \Sigma)$ is Joint Gaussian

当 $n=1$ 时, $X \sim N(\mu, \sigma^2)$, $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

当 $n=2$ 时, $(x_1, x_2) \sim N(\mu_1, \mu_2, \sigma_1, \sigma_2, \rho)$,

$$f_{x_1, x_2}(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)}\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2 - 2\rho\left(\frac{x_1-\mu_1}{\sigma_1}\right)\left(\frac{x_2-\mu_2}{\sigma_2}\right)\right]$$

现考虑 n 维: $X \in \mathbb{R}^n$, $X \sim N(\mu, \Sigma)$

$$f_X(x) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right]$$

其中 $E(X) = \mu$, $\Sigma = E(X-\mu)(X-\mu)^T$ (Covariance Matrix)

下验证 $f_X(x)$ 为密度: ① $f_X(x) \geq 0$

$$\textcircled{2} \int_{\mathbb{R}^n} f_X(x) dx = 1$$

Pf: $\int_{\mathbb{R}^n} f_X(x) dx = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \int_{\mathbb{R}^n} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right] dx \quad \underbrace{dx_1 \dots dx_n}_{\text{...}} \quad \textcircled{1}$

由于 $\Sigma = \Sigma^T \geq 0$, 因此可以进行正交分解: $\Sigma = U^T \Lambda U$, $\Lambda = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$

则: $\Sigma^{-1} = U^T \Lambda^{-1} U$, $\Lambda^{-1} = \text{diag}\left(\frac{1}{\sigma_1^2}, \dots, \frac{1}{\sigma_n^2}\right)$.

作变换: $y = U(x-\mu)$

$$\text{则 } \textcircled{1} = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2} y^T \Lambda^{-1} y\right) dy$$

$$\quad \quad \quad = \prod_{k=1}^n \sigma_k$$

$$= \frac{1}{(\sqrt{2\pi})^n \prod_{k=1}^n \sigma_k} \int_{\mathbb{R}^n} \exp\left(-\sum_{k=1}^n \frac{y_k^2}{2\sigma_k^2}\right) dy$$

$$= \prod_{k=1}^n \frac{1}{\sqrt{2\pi} \sigma_k} \int_{\mathbb{R}} \exp\left(-\frac{y_k^2}{2\sigma_k^2}\right) dy_k$$

$$= \prod_{k=1}^n 1 = 1 \quad \square$$

Characteristic Function

$$\phi_X(w) = E(\exp(jw^T x)), \quad x \in \mathbb{R}^n$$

看一下 n 维高斯的 $\phi_X(w)$:

$$\phi_X(w) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \int_{\mathbb{R}^n} \exp(jw^T x) \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right) dx \quad \dots \textcircled{2}$$

$$\text{不妨考虑: } -\frac{1}{2\sigma^2}(x-\mu)^2 + jwx = -\frac{1}{2\sigma^2}(x-(\mu+j\sigma^2 w))^2 + jw\mu - \frac{1}{2}\sigma^2 w^2$$

$$\text{可以有: } -\frac{1}{2}(x-\mu-j\Sigma w)^T \Sigma^{-1} (x-\mu-j\Sigma w) + jw^T \mu - \frac{1}{2} w^T \Sigma w$$

$$\text{则 } \textcircled{2} = \exp(jw^T \mu - \frac{1}{2} w^T \Sigma w) \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}(x-\mu-j\Sigma w)^T \Sigma^{-1} (x-\mu-j\Sigma w)\right) dx$$

$$= \exp(jw^T \mu - \frac{1}{2} w^T \Sigma w)$$

因此 $\phi_x(w) = \exp(jw^T \mu - \frac{1}{2} w^T \Sigma w)$

Linearity : 线性性

若 $X \in \mathbb{R}^n$, 且 $X \sim N(\mu, \Sigma)$. 有 $A \in \mathbb{R}^{m \times n}$, $Y = AX$

一定有 $Y \sim N(A\mu, A\Sigma A^T)$

$$\begin{aligned} \text{Pf: } \phi_Y(w) &= E(\exp(jw^T Y)) \\ &= E(\exp(jw^T AX)) = E(\exp(j(A^T w)^T X)) \\ &= \phi_X(A^T w) \\ &= \exp(j(A^T w)^T \mu - \frac{1}{2} (A^T w)^T \Sigma (A^T w)) \\ &= \exp(jw^T (A\mu) - \frac{1}{2} w^T (A\Sigma A^T) w) \end{aligned}$$

故 $Y \sim N(A\mu, A\Sigma A^T)$

□

→ 高斯的边缘分布一定为高斯. 反之不成立

反例: $f_{x_1, x_2}(x_1, x_2) = \frac{1}{2\pi} \exp(-\frac{x_1^2 + x_2^2}{2}) + c(x, y)$

$$\int_{\mathbb{R}} c(x, y) dx = \int_{\mathbb{R}} c(x, y) dy = 0$$

$$\text{可取 } c(x, y) = \sin x \sin y$$

$$\text{最终可取 } f_{x_1, x_2}(x_1, x_2) = \frac{1}{2\pi} \exp(-\frac{x_1^2 + x_2^2}{2}) (1 + \sin x \sin y)$$

Property :

① $x_1, \dots, x_n$ 独立. $x_k \sim N(\mu_k, \sigma_k^2) \Rightarrow (x_1, \dots, x_n) \sim N(\mu, \Sigma)$

$$\text{其中 } \mu = (\mu_1, \dots, \mu_n)^T, \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$$

② $(x_1, \dots, x_n)^T = X \sim N(\mu, \Sigma) \iff \forall \alpha \in \mathbb{R}^n, \alpha^T X \sim N(\alpha^T \mu, \alpha^T \Sigma \alpha)$

Pf: " $\Rightarrow$ " 显然

$$\begin{aligned} \text{"}\Leftarrow\text{" } \forall \alpha, \phi_{\alpha^T X}(w) &= E(\exp(jw \alpha^T X)) \\ \text{取 } w &= 1, \phi_{\alpha^T X}(1) = E(\exp(j \alpha^T X)) = \phi_X(\alpha) \\ &= \exp(j \alpha^T \mu - \frac{1}{2} \alpha^T \Sigma \alpha) \end{aligned}$$

故 $X \sim N(\mu, \Sigma)$

□

Diffusion (DDPM) Denoise Diffusion Probability Model

当作一幅画, 计算机从联合分布采样 (Sampling). Computer Drawing $\iff$ Random Number Sampling

e.g. 画一座山.  $x_0 \xrightarrow{+N_1} x_1 \xrightarrow{+N_2} x_2 \xrightarrow{+N_3} x_3 \longrightarrow \dots \longrightarrow x_T \sim N(0, \sigma^2 I)$

$$x_0 \xleftarrow{NN} \dots \xleftarrow{NN} x_{t-1} \xleftarrow{NN} x_t \xleftarrow{NN} \dots \xleftarrow{NN} x_T$$

Reverse Inference

Property :

$$\begin{aligned}
 ① \quad x_t &= \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t} \varepsilon, \quad \varepsilon \sim N(0,1) \\
 &= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1-\alpha_{t-1}} \varepsilon_1 \right) + \sqrt{1-\alpha_t} \varepsilon_2 \\
 &= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \underbrace{\sqrt{\alpha_t} \sqrt{1-\alpha_{t-1}} \varepsilon_1 + \sqrt{1-\alpha_t} \varepsilon_2}_{\leftarrow} \\
 &= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{1-\alpha_t \alpha_{t-1}} \varepsilon \\
 &\vdots \\
 &= \sqrt{\alpha_t \alpha_{t-1} \cdots \alpha_1} x_0 + \sqrt{1-\alpha_t \alpha_{t-1} \cdots \alpha_1} \varepsilon \\
 &\triangleq \sqrt{\alpha_t} x_0 + \sqrt{1-\alpha_t} \varepsilon
 \end{aligned}$$

$$\begin{aligned}
 ② \quad P(x_{t-1} | x_t, x_0) \\
 = P(x_t | x_{t-1}, x_0) \frac{P(x_{t-1} | x_0)}{P(x_t | x_0)} \quad \text{Bayes' Formula}
 \end{aligned}$$

其中 $P(x_t | x_{t-1}, x_0) \sim N(\sqrt{\alpha_t} x_{t-1}, (1-\alpha_t)I)$

$$P(x_t | x_0) \sim N(\sqrt{\alpha_t} x_0, (1-\alpha_t)I), \quad P(x_{t-1} | x_0) \sim N(\sqrt{\alpha_{t-1}} x_0, (1-\alpha_{t-1})I)$$

考虑二次型：

$$\begin{aligned}
 &-\frac{1}{2} \left(\frac{(x_t - \sqrt{\alpha_t} x_{t-1})^2}{1-\alpha_t} + \frac{(x_{t-1} - \sqrt{\alpha_{t-1}} x_0)^2}{1-\alpha_{t-1}} - \frac{(x_t - \sqrt{\alpha_t} x_0)^2}{1-\alpha_t} \right) \quad \text{瞄准 } x_{t-1} \text{ 配方, } x_t, x_0 \text{ 为常数} \\
 &= -\frac{1}{2} \left(\frac{x_t^2 - 2\sqrt{\alpha_t} x_t x_{t-1} + \alpha_t x_{t-1}^2}{1-\alpha_t} + \frac{x_{t-1}^2 - 2\sqrt{\alpha_{t-1}} x_{t-1} x_0 + \alpha_{t-1} x_0^2}{1-\alpha_{t-1}} - \frac{x_t^2 - 2\sqrt{\alpha_t} x_t x_0 + \alpha_t x_0^2}{1-\alpha_t} \right)
 \end{aligned}$$

$$= -\frac{1}{2} \left(\left(\frac{\alpha_t}{1-\alpha_t} + \frac{1}{1-\alpha_{t-1}} \right) x_{t-1}^2 - 2 \left(\frac{\sqrt{\alpha_t}}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_{t-1}} x_0 \right) x_{t-1} + C(x_0, x_t) \right)$$

$$\begin{aligned}
 \text{Var} &= \left(\frac{\alpha_t}{1-\alpha_t} + \frac{1}{1-\alpha_{t-1}} \right)^{-1} = \left(\frac{\alpha_t (1-\alpha_{t-1}) + (1-\alpha_t)}{(1-\alpha_t)(1-\alpha_{t-1})} \right)^{-1} \\
 &= \left(\frac{1-\alpha_t}{(1-\alpha_t)(1-\alpha_{t-1})} \right)^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Mean} &= \frac{\frac{\sqrt{\alpha_t}}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}}}{1-\alpha_{t-1}} x_0}{\frac{1-\alpha_t}{(1-\alpha_t)(1-\alpha_{t-1})}} = \frac{\sqrt{\alpha_t} (1-\alpha_{t-1})}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}} (1-\alpha_t)}{1-\alpha_t} \boxed{x_0} \quad \text{Ground-Truth}
 \end{aligned}$$

③ 将上式的 x_0 替换为 ε

$$\begin{aligned}
 \frac{\sqrt{\alpha_t} (1-\alpha_{t-1})}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}} (1-\alpha_t)}{1-\alpha_t} \boxed{x_0} &= \frac{\sqrt{\alpha_t} (1-\alpha_{t-1})}{1-\alpha_t} x_t + \frac{\sqrt{\alpha_{t-1}} (1-\alpha_t)}{1-\alpha_t} \left(\frac{1}{\sqrt{\alpha_t}} (x_t - \sqrt{1-\alpha_t} \varepsilon) \right) \\
 &= \left(\frac{\sqrt{\alpha_t} (1-\alpha_{t-1})}{1-\alpha_t} + \frac{\sqrt{\alpha_{t-1}} (1-\alpha_t)}{(1-\alpha_t) \sqrt{\alpha_t}} \right) x_t - \frac{\sqrt{\alpha_{t-1}} (1-\alpha_t)}{(1-\alpha_t) \sqrt{\alpha_t}} \varepsilon
 \end{aligned}$$

$$= \frac{\alpha_t \sqrt{\bar{\alpha}_{t-1}} \left(\sqrt{\alpha_t} \sqrt{\bar{\alpha}_t} (1 - \bar{\alpha}_{t-1}) + \sqrt{\bar{\alpha}_{t-1}} (1 - \alpha_t) \right)}{(1 - \bar{\alpha}_t) \sqrt{\bar{\alpha}_t}} x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t} \sqrt{\alpha_t}} \varepsilon$$

$$= \frac{\sqrt{\bar{\alpha}_{t-1}} \left(\alpha_t (1 - \bar{\alpha}_{t-1}) + 1 - \alpha_t \right)}{(1 - \bar{\alpha}_t) \sqrt{\bar{\alpha}_t}} = \frac{\sqrt{\bar{\alpha}_{t-1}} (1 - \bar{\alpha}_t)}{(1 - \bar{\alpha}_t) \sqrt{\bar{\alpha}_t}} = \frac{1}{\sqrt{\alpha_t}}$$

$$= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon \right)$$